

Dohyeop Lim

Undergraduate Student in Applied Artificial Intelligence

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Education

Seoul National University of Science and Technology 2024 – Present
B.Eng. in Applied Artificial Intelligence Seoul, Korea
GPA: 4.34 / 4.5 · 1st in Department

Research Experience

Visual Intelligence Lab, SeoulTech Dec. 2025 – Present
Undergraduate Researcher (*Advisor: Prof. Jongyoul Park*) Seoul, Korea

- Conduct research on cross-document consistency verification and structured identifier recognition for a national R&D project with KETI on automated data preprocessing for industrial AI.
- Develop synthetic document generation methods with dynamic table layouts, structured form templates, and automated tagging for document understanding research.
- Led an 8,000-image annotation project for evaluation dataset construction, including annotator recruitment and deployment of a custom labeling tool.

Publications

E-ACT: Enforcing Arithmetic Constraints in CTC Trellises for Structured Identifier Recognition
Dohyeop Lim, MinKi Jeong, Jongyoul Park. • WACV 2027, under review.

- Developed a factorized CTC decoder that enforces identifier length, allowed characters, and checksum rules during decoding without retraining.
- Improved exact-match accuracy by 12.3–31.2 pp across four benchmarks and prevented 89.5–93.6% of single-edit errors in unconstrained OCR outputs from propagating to final predictions.

Imagine the Structure Before You Speak: Structural Imagination in Latent Space for Table Recognition
MinKi Jeong, Dohyeop Lim, Jongyoul Park. • WACV 2027, under review.

Selected Projects

Collog  Jul. – Aug. 2026
Turns everyday family calls into actionable health and lifestyle insights for older adults.

- Built a pipeline for transcription, health information extraction, and acoustic analysis from family call audio.
- Tracked changes across calls and generated questions for future conversations based on previous records.

MochiCall  Jul. 2026
Converts spoken requests for cross-regional hospital transportation into dispatch-ready trip records.

- Fine-tuned and evaluated domain-specific Korean ASR models using LoRA.
- Combined specialized and multi-domain adapters to improve transcription under domain shift.

WONNIT  Jul. – Sep. 2025
Supports reliable short-term rentals of idle spaces through automated visual verification.

- Developed models for scene classification, object detection, and before-and-after change detection.
- Converted and deployed the vision models with Core ML for on-device inference on iOS.

Awards & Honors

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- Academic Excellence Scholarship, SeoulTech 2024 – 2026
Full tuition for five consecutive semesters
 - Honorable Mention (Overall), 14th LIKELION Univ. Central Hackathon Aug. 2026
Selected among 317 teams with Collog
 - 2nd Place (AAC Track), 14th LIKELION Univ. Central Hackathon Aug. 2026
Ranked 2nd out of 128 teams
 - Grand Prize, 2026 AICOSS Summer AI SCI Bootcamp Competition Jul. 2026
Awarded for MochiCall

Skills & Languages

Programming	Python, Swift, TypeScript
Machine Learning	PyTorch, OpenCV
Mobile	SwiftUI
Web & Tools	Next.js, Docker, Git, $\text{\LaTeX}$
Spoken Languages	Korean (Native), English (TOEFL iBT 100/120, Aug. 2026)

Additional Experience

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- LIKELION University 2025 – Present
Vice President, 14th Cohort; Member, 13th Cohort, SeoulTech Chapter
Coordinated a 30-member chapter, organized four technical sessions, invited talks, and hackathon programs.
 - Google Developer Groups on Campus 2025 – Present
Core Team Member, 5th and 6th Cohorts, SeoulTech Chapter
Organized 13 technical sessions for the campus developer community during the 5th cohort.
 - Visiting Student, Technische Hochschule Ulm, Germany Jan. – Feb. 2026
Completed coursework in Deep Learning for Industrial Data.
Developed **DocFusionX** , a structure-aware GraphRAG system for reliable reasoning over long documents.